MSG No. 133.460 $P4_2/nbc1'$ [Type II, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `222.1'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,4,5,6,17,20,21,22]`
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[2,3,7,8,18,19,23,24]`
3	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[9,12,13,14,25,28,29,30]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[10,11,15,16,26,27,31,32]`

Table 2: Wyckoff site: `4b`, site symmetry: `222.1'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,4,5,6,17,20,21,22]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[2,3,7,8,18,19,23,24]`
3	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[9,12,13,14,25,28,29,30]`
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[10,11,15,16,26,27,31,32]`

Table 3: Wyckoff site: `4c`, site symmetry: `2.221'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,6,7,8,17,22,23,24]`
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[2,3,4,5,18,19,20,21]`
3	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[9,14,15,16,25,30,31,32]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[10,11,12,13,26,27,28,29]`

Table 4: Wyckoff site: `4d`, site symmetry: `-4..1'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,6,10,11,17,22,26,27]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,3,9,14,18,19,25,30]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[4,5,15,16,20,21,31,32]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[7,8,12,13,23,24,28,29]`

Table 5: Wyckoff site: `8e`, site symmetry: `-11'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,9,17,25]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,10,18,26]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,11,19,27]`
4	$\left[0, \frac{1}{2}, 0\right]$	`[4,12,20,28]`
5	$\left[\frac{1}{2}, 0, 0\right]$	`[5,13,21,29]`
6	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[6,14,22,30]`
7	$\left[0, 0, \frac{1}{2}\right]$	`[7,15,23,31]`

8	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[8,16,24,32]

Table 6: Wyckoff site: 8f, site symmetry: 2..1'

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	[1,6,17,22]
2	$\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[2,3,18,19]
3	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	[4,5,20,21]
4	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[7,8,23,24]
5	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	[9,14,25,30]
6	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[10,11,26,27]
7	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	[12,13,28,29]
8	$\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[15,16,31,32]

Table 7: Wyckoff site: 8g, site symmetry: 2..1'

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	[1,6,17,22]
2	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[2,3,18,19]
3	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	[4,5,20,21]
4	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[7,8,23,24]
5	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	[9,14,25,30]
6	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[10,11,26,27]
7	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	[12,13,28,29]
8	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[15,16,31,32]

Table 8: Wyckoff site: 8h, site symmetry: .2.1'

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ 0\right]$	[1,4,17,20]
2	$\left[\frac{1}{4},\ x,\ \frac{1}{2}\right]$	[2,7,18,23]
3	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[3,8,19,24]
4	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ 0\right]$	[5,6,21,22]
5	$\left[-x,\ \frac{3}{4},\ 0\right]$	[9,12,25,28]
6	$\left[\frac{3}{4},\ -x,\ \frac{1}{2}\right]$	[10,15,26,31]
7	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[11,16,27,32]
8	$\left[x+\frac{1}{2},\ \frac{3}{4},\ 0\right]$	[13,14,29,30]

Table 9: Wyckoff site: 8i, site symmetry: .2.1'

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{2}\right]$	[1,4,17,20]
2	$\left[\frac{1}{4},\ x,\ 0\right]$	[2,7,18,23]
3	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ 0\right]$	[3,8,19,24]

continued ...

Table 9

No.	position	mapping
4	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{2}\right]$	[5,6,21,22]
5	$\left[-x,\ \frac{3}{4},\ \frac{1}{2}\right]$	[9,12,25,28]
6	$\left[\frac{3}{4},\ -x,\ 0\right]$	[10,15,26,31]
7	$\left[\frac{3}{4},\ x+\frac{1}{2},\ 0\right]$	[11,16,27,32]
8	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}\right]$	[13,14,29,30]

Table 10: Wyckoff site: 8j, site symmetry: ..21'

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{4}\right]$	[1,7,17,23]
2	$\left[\frac{1}{2}-x,\ x,\ \frac{3}{4}\right]$	[2,5,18,21]
3	$\left[x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[3,4,19,20]
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[6,8,22,24]
5	$\left[-x,\ -x,\ \frac{3}{4}\right]$	[9,15,25,31]
6	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	[10,13,26,29]
7	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[11,12,27,28]
8	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[14,16,30,32]

Table 11: Wyckoff site: 16k, site symmetry: 11'

No.	position	mapping
1	$[x,\ y,\ z]$	[1,17]
2	$\left[\frac{1}{2}-y,\ x,\ z+\frac{1}{2}\right]$	[2,18]
3	$\left[y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[3,19]
4	$\left[x,\ \frac{1}{2}-y,\ -z\right]$	[4,20]
5	$\left[\frac{1}{2}-x,\ y,\ -z\right]$	[5,21]
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[6,22]
7	$\left[y,\ x,\ \frac{1}{2}-z\right]$	[7,23]
8	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[8,24]
9	$[-x,\ -y,\ -z]$	[9,25]
10	$\left[y+\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$	[10,26]
11	$\left[-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[11,27]
12	$\left[-x,\ y+\frac{1}{2},\ z\right]$	[12,28]
13	$\left[x+\frac{1}{2},\ -y,\ z\right]$	[13,29]
14	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[14,30]
15	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	[15,31]
16	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[16,32]